Finite Fourier upper bound: normalization checked

Result. Lean proves the finite-abelian Upper Bound Lemma for actual PMFs and a genuine uniform law: half-L1 distance squared is at most one quarter of the nontrivial-character energy. Finite complex Plancherel is derived from actual character orthogonality, then combined with squared Cauchy–Schwarz and actual probability normalization. No spectral-energy or mixing bound is supplied as a premise. [pcc:unverified]

Derived Parseval	Actual finite complex character transform Sum of squared Fourier norms = |G|·sum of squared norms
Actual probability	Half-L1 = maximum event discrepancy; attaining event proved Half-L1² ≤ nontrivial-character energy / 4
Sharp control	ZMod2 pointmass/uniform: fullL1=1, halfL1=1/2, energy=1 Quarter bound is equality for halfL1; false for fullL1

Actual source distance and uniform law. FullL1 sums absolute differences of actual point probabilities; source distance is halfL1. Lean proves its range[0,1], every-event bound and an explicit positive-difference event attaining it, so it equals maximum event discrepancy. Actual uniform point masses are1/|G|, event probabilities |A|/|G| and character coefficients1/0 at trivial/nontrivial modes. Probability normalization removes the trivial mode. [pcc:unverified]

Actual independent walk. The preceding actual PMF character-moment law gives halfL1(walk,uniform)² ≤ one quarter of the sum over nontrivial characters of normSq($\Sigma_i q_i \psi(a_i)$)^t, for every finite abelian group, time and start. The same right side bounds every event's squared discrepancy from |A|/|G|. No supplied convolution-power or spectral formula is used. [pcc:unverified]

Normalization counterexample and sharpness. Actual ZMod2 pointmass0 versus uniform has fullL1=1, halfL1=1/2 and nontrivial energy1. Lean directly refutes using the quarter factor with fullL1, proves equality with halfL1 and shows any universal halfL1 constant on this group must be at least1/4. Original1981 uses fullL1; author1987 and Hildebrand use halfL1. Their qualified statements are retained. [pcc:unverified]

Primary review. Diaconis author report269, April1987: selected chapter3B printed29–33/PDF35–39 OCR read; visual29/32/33/contents, primary hash checked; not the1988 book pagination or full271-page read. Published Diaconis–Shahshahani1981: selected normalization/convolution/Plancherel and full Lemma14 proof168 read; visual159/161/162/168, not full21-page read. Hildebrand1994 full13-page review credited and195 rechecked; later preprint selected4.1/4.2 only. Raw primary PDFs/scans/OCR remain local, metadata retained. [pcc:unverified]

Checks and remainder. Whole gate: 38 declarations/0 failures/0 trusted native exceptions. All 99 build/provenance files retained losslessly in 40 objects, including first gate and dry-binding correction; final five gate files and recursive pins. Lean4.35.0/Mathlib56a00585…, standard foundations/default limits. Signed standing unverified. Nonabelian matrix/measure-tvDist interfaces and random-support spectral averaging remain open. No holes/custom axioms/native evaluation/raised limits/global flags/waivers/native experiment/spend/upload. [pcc:unverified]